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# ***Past Paper 4***

Time: 25 minutes | Paper Reference **PMAT2/N04**

# Mathematics

## Level 2
## Section A (Non-Calculator)



**You must have:**
Pen, HB pencil, eraser, ruler graduated in cm and mm, protractor, pair of compasses. Tracing paper may be used.

Total Marks
16 /16

**My signature confirms that I will not discuss the content of the test with anyone.**

**Signature:** Master Functional Skills (MFS)

## Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Sign the declaration.
- Answer **all** questions.
- Write your final answers in the boxes provided.
- Answer the questions in the spaces provided – *there may be more space than you need.*
- You **must** show clearly how you get your answers in the spaces provided. Marks will be awarded for your working out.
- Check your working and answers at each stage.
- Diagram are **not** accurately drawn, unless otherwise indicated.
- **Calculators may not be used**.
- Take the value of $\pi$ to be 3.14

## Information

- The total mark for this section is 16.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*
- This sign ☑ shows where marks will be awarded for showing your checks.

## Advice

- Read each question carefully before you start to answer it.
- Check your answers if you have time at the end.

*Turn over* ▶

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## SECTION A

**Answer ALL questions. Write your answers in the spaces provided.**

**1** Shannon works in a mobile phone shop.

This week Shannon makes

- 19 pay as you go phone sales
- 57 SIM card with handset contract sales.

The remaining sales were SIM card only contracts.

Shannon made a total of 114 sales during this week.

What fraction of the sales made by Shannon this week were SIM card only contracts?
Write your fraction in its simplest form.

**(3)**

Step 1: Calculate total known contract sales

$$\begin{array}{rr} & {\scriptstyle 1\phantom{0}} \\ & 1\ 9 \\ + & 5\ 7 \\ \hline & 7\ 6 \end{array}$$

(carry: 9 + 7 = 16)

PAYG + handset sales

Step 2: Subtract from total sales to find SIM-only contracts

$$\begin{array}{rr} & {\scriptstyle 0\ \ 10\ \ 14} \\ & \not{1}\ \not{1}\ \not{4} \\ - & 7\ 6 \\ \hline & 3\ 8 \end{array}$$

SIM-only contracts [M1]

Step 3: Express as a fraction of all sales (114) and simplify

$$\frac{38}{114} \begin{matrix} \div 2 \\ \div 2 \end{matrix} = \frac{19}{57} \begin{matrix} \div 19 \\ \div 19 \end{matrix} = \frac{1}{3}$$

(or divide by HCF = 38) [M1 A1]

1/3

**(Total for Question 1 is 3 marks)**

**2** Grace and James share £72 in the ratio 4 : 5

(a) Work out the amount of money that James gets.
You **must** show your working.

**(2)**

*Step 1: Find total ratio parts*

**4 + 5 = 9 parts in total**

*Step 2: Calculate the value of 1 part*

**£72 ÷ 9 = £8 per part [M1]**

*Step 3: Calculate James's share (5 parts)*

**James = 5 × £8 = £40 [A1]**

*(Grace gets 4 × £8 = £32)*

£ **40**

☑ (b) Show a check of your working.

**(1)**

*Reverse check using division and multiplication:*

**40 ÷ 5 = 8**

**8 × 9 = 72** ✓

*Alternatively: £32 + £40 = £72 [B1]*

**(Total for Question 2 is 3 marks)**

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**3** Brian collects loyalty points at the supermarket.
The total value of the loyalty points he has collected is £17.50

Brian has this information.

> Exchange loyalty points for travel vouchers and get 3 times the value

Travel vouchers can be used to pay for ferry tickets.

Brian wants to book 2 ferry tickets.
The ferry tickets cost £33.25 and £29

Brian will exchange all his loyalty points for travel vouchers.
He will use all the travel vouchers towards the cost of the ferry tickets.

Brian will pay for the rest of the cost of the ferry tickets with his bank card.

How much will Brian pay with his bank card?

**(4)**

*Step 1: Total cost of 2 ferry tickets*

$$\begin{array}{r} {\scriptstyle 1\phantom{00}} \\ 33.25 \\ +\ 29.00 \\ \hline 62.25 \end{array}$$

*Total cost = £62.25 [M1]*

*Step 2: Value of travel vouchers (3 times points value)*

$3 \times £17.50 = £52.50$ **[M1]** *(3 × £17 = £51, 3 × £0.50 = £1.50)*

*Step 3: Remaining balance paid with bank card*

$$\begin{array}{r} {\scriptstyle 5\ 11\ 12} \\ \cancel{6}\ \cancel{2}.\cancel{2}5 \\ -\ 52.50 \\ \hline 9.75 \end{array}$$

*Balance to pay = £9.75 [M1 A1]*

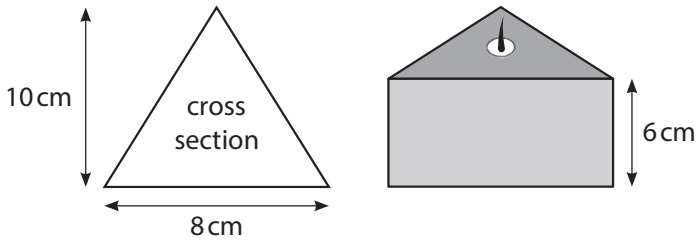
£ 9.75

**(Total for Question 3 is 4 marks)**

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**4** Zia makes and sells candles.
She wants to make candles in the shape of a triangular prism.

Zia has this sketch of the triangular face and the candle.




Zia will melt wax to make the candles.
She knows that

- each 1 kg of solid wax makes 1.2 litres of melted wax
- 1 litre = 1000 $cm^3$
- the volume of a prism = area of cross section × height

Zia has a 5 kg bag of solid wax.
She thinks that she can make at least 20 candles with this bag of solid wax.

Is Zia correct?
Show why you think this.

**(6)**

*Step 1: Area of triangular cross-section*

Area = ½ × base × height

= ½ × 8 × 10 = 40 $cm^2$ [M1]

*Step 2: Volume of 1 candle (triangular prism)*

Volume = area of cross-section × length

= 40 × 6 = 240 $cm^3$ [M1]

*Step 3: Total wax volume needed for 20 candles*

Total volume needed = 20 × 240 $cm^3$

= 4800 $cm^3$ [M1]

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Step 4: Melted wax available from 5 kg bag

**5 × 1.2 litres = 6.0 litres of melted wax [M1]**

Step 5: Convert litres to $cm^3$ (1 litre = 1000 $cm^3$)

**6.0 × 1000 = 6000 $cm^3$ of wax available [M1]**

Step 6: Compare wax available vs wax needed

**Wax available = 6000 $cm^3$**

**Wax needed for 20 candles = 4800 $cm^3$**

**6000 $cm^3$ > 4800 $cm^3$ (1200 $cm^3$ extra wax)**

Alternative check: 6000 ÷ 240 = 25 candles can be made (25 ≥ 20)

**Conclusion: Yes, Zia is correct.**

She has 6000 $cm^3$ of melted wax, enough to make 25 candles.
This is more than the 20 candles needed (4800 $cm^3$). [A1]

**Yes (6000 $cm^3$ > 4800 $cm^3$)**

**(Total for Question 4 is 6 marks)**

**TOTAL FOR SECTION A IS 16 MARKS**

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# ***Past Paper 4***

Time: 1 hour 30 minutes | Paper Reference **PMAT2/C04**

# Mathematics

## Level 2
## Section B (Calculator)



**You must have:**
Pen, HB pencil, eraser, ruler graduated in cm and mm, protractor, pair of compasses. Tracing paper may be used.

Total Marks 48 /48

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### Instructions

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- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Sign the declaration.
- Answer **all** questions.
- Write your final answers in the boxes provided.
- Answer the questions in the spaces provided – *there may be more space than you need.*
- You **must** show clearly how you get your answers in the spaces provided. Marks will be awarded for your working out.
- Check your working and answers at each stage.
- Diagram are **not** accurately drawn, unless otherwise indicated.
- **Calculators may be used**.
- If your calculator does not have a $\pi$ button take the value of $\pi$ to be 3.14

### Information

- The total mark for this section is 48.
- The total mark for this paper is 64.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*
- This sign ☑ shows where marks will be awarded for showing your checks.

### Advice

- Read each question carefully before you start to answer it.
- Check your answers if you have time at the end.

*Turn over* ▶

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## SECTION B

**Answer ALL questions. Write your answers in the spaces provided.**

**1** Johan is cooking using an old recipe.
The recipe says to preheat the oven to a temperature of 350 °F.

The oven Johan uses is marked in centigrade (°C).
He finds this formula to change from °F to °C.

$$C = \frac{5(F - 32)}{9}$$

$C$ = temperature °C
$F$ = temperature °F

Johan sets the oven temperature to 190 °C.

Has Johan set the oven to the correct temperature for this recipe?

**(3)**

Step 1: Substitute F = 350 °F into the formula

C = 5 × (350 − 32) ÷ 9 [M1]

Step 2: Calculate inside brackets, then multiply and divide

350 − 32 = 318

C = (5 × 318) ÷ 9 = 1590 ÷ 9 [M1]

C = 176.666... °C ≈ 176.7 °C (or 177 °C)

Reverse check: 190 °C in °F = (190 × 9 ÷ 5) + 32 = 342 + 32 = 374 °F

**Conclusion: No, Johan has not set the oven correctly.**

350 °F is approximately 177 °C (176.7 °C).
Johan set it to 190 °C, which is 13.3 °C too hot (or 374 °F > 350 °F). [A1]

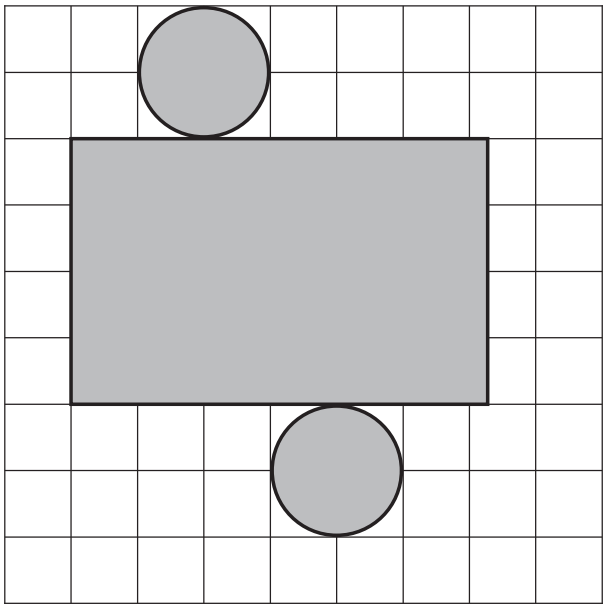
**No (176.7 °C)**

**(Total for Question 1 is 3 marks)**

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**2** Here is the net of a 3D object.

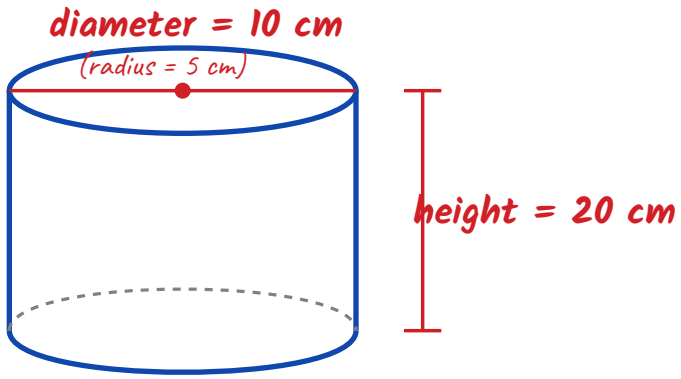


**Key** 1 cm on the grid represents 5 cm on the 3D object

Draw a sketch of the 3D object.
Remember to label the dimensions on your sketch.

**(3)**

Scale: 1 cm on grid = 5 cm on 3D object --> 3D object is a CYLINDER [M1]

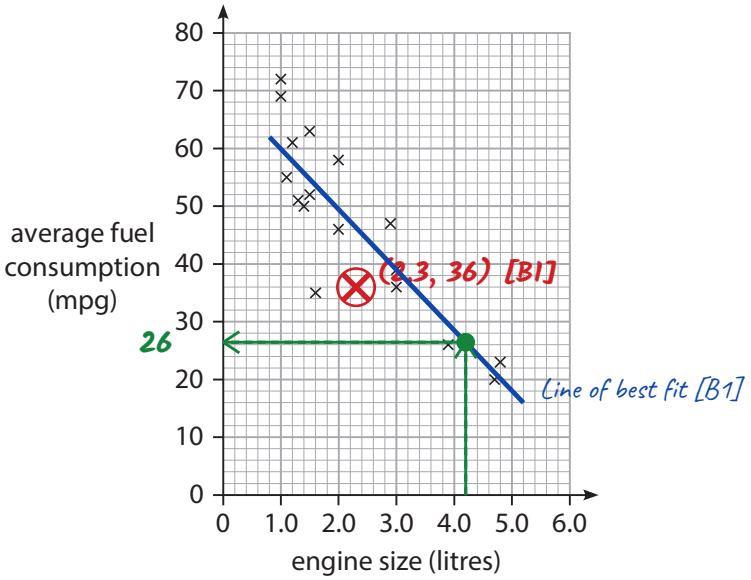



- Circles on net: diameter = 2 cm × 5 = 10 cm (radius = 5 cm) [B1]
- Rectangle on net: height = 4 cm × 5 = 20 cm [B1]

Accurate 3D cylinder sketch with diameter 10 cm and height 20 cm [3/3]

**(Total for Question 2 is 3 marks)**

**3** The scatter diagram shows some information about the engine size in litres and average fuel consumption in miles per gallon (mpg) of some cars.




Here is the information for another car

- engine size 2.3 litres, average fuel consumption 36 mpg.

(a) Plot this information on the scatter graph.

**(1)**

(b) Draw a line of best fit on the scatter graph.

**(1)**

Mikael buys a car with an engine size of 4.2 litres.

(c) Use your line of best fit to estimate the average fuel consumption of this car.

**(1)**

Read from line of best fit at 4.2 litres = 26 mpg [B1]

**26** mpg

(d) What type of correlation is shown in this scatter diagram?
Tick [✔] a box to show your answer.

**(1)**

| unlikely | negative | even | neutral | positive | likely |
|---|---|---|---|---|---|
| ☐ | ☑ | ☐ | ☐ | ☐ | ☐ |

Negative [B1]

**(Total for Question 3 is 4 marks)**

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**4** (a) Work out $\dfrac{(3^3 - 4^2) + 7}{2.5}$

**(1)**

*Step 1: Calculate powers (indices)*

$3^3 = 3 \times 3 \times 3 = 27$

$4^2 = 4 \times 4 = 16$

*Step 2: Calculate inside brackets, then add 7 and divide by 2.5*

$(27 - 16) + 7 = 11 + 7 = 18$

$18 \div 2.5 = 7.2$ [A1]

7.2

(b) Write the following values in order of size.
Start with the smallest value.

0.5 $\quad \frac{5}{9} \quad$ 0.53 $\quad \frac{4}{7} \quad$ 47%

**(2)**

*Convert all numbers to decimals to compare:*

$47\% = 47 \div 100 = 0.470$

$0.5 = 0.500$

$0.53 = 0.530$

$5/9 = 5 \div 9 = 0.5555... \approx 0.556$

$4/7 = 4 \div 7 = 0.5714... \approx 0.571$ [M1]

*Order from smallest to largest:*

$0.470 < 0.500 < 0.530 < 0.556 < 0.571$

47% , 0.5 , 0.53 , 5/9 , 4/7 [A1]

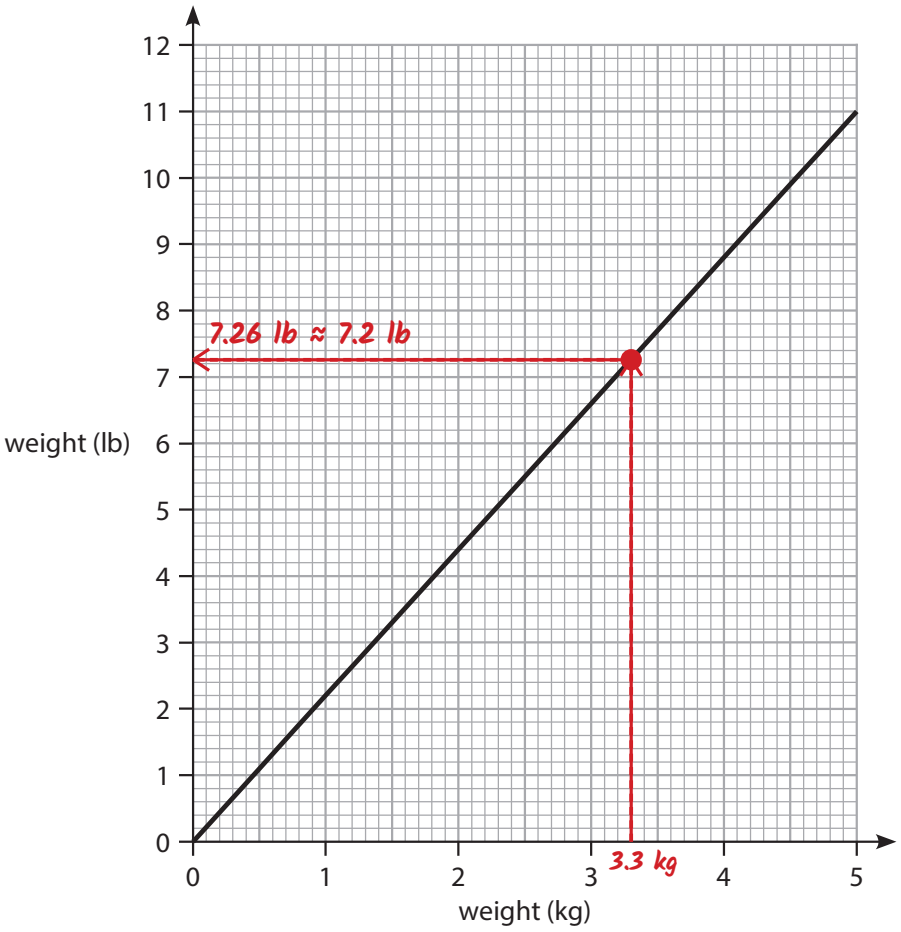
**(Total for Question 4 is 3 marks)**

**5** Misbah is a midwife at her local hospital.
She is writing a report about changes in the average weight of a baby at birth.

Misbah has this information about birth weights in 2018 at the hospital.

| **Weight (kg)** | 2.9 | 3.0 | 3.1 | 3.2 | 3.3 | 3.4 | 3.5 | 3.6 | 3.7 |
|---|---|---|---|---|---|---|---|---|---|
| **Frequency** | 26 | 48 | 74 | 113 | 167 | 132 | 109 | 92 | 36 |

Misbah can use this graph to change between kg and lb. Mode = 3.3 kg (freq 167) [B1]




Misbah wants to write about the percentage change in the modal weight of a baby from 1998 to 2018

She finds out that the **modal** weight of a baby in 1998 was 7.7 lb

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What is the percentage change in the modal weight of a baby from 1998 to 2018?
Write your answer to the nearest whole number.

**(5)**

Step 1: Modal weight in 2018 = 3.3 kg (highest frequency = 167) [B1]

Step 2: Read conversion graph: 3.3 kg = 7.26 lb ≈ 7.2 lb (or 7.7 lb = 3.5 kg) [M1]

Step 3: Formula for percentage change: ((New – Old) ÷ Old) × 100%

**Using weights in pounds (lb):**

**Change = 7.2 – 7.7 = –0.5 lb (decrease) [M1]**

**Percentage Change = (–0.5 ÷ 7.7) × 100%**

**= –0.06493... × 100% = –6.49% [M1]**

Alternative check using kilograms (kg):

Change = 3.3 – 3.5 = –0.2 kg; (–0.2 ÷ 3.5) × 100% = –5.71%

Step 4: Round to the nearest whole number

**–6.49% (or –5.71%) rounds to 6% decrease (or –6%) [A1]**

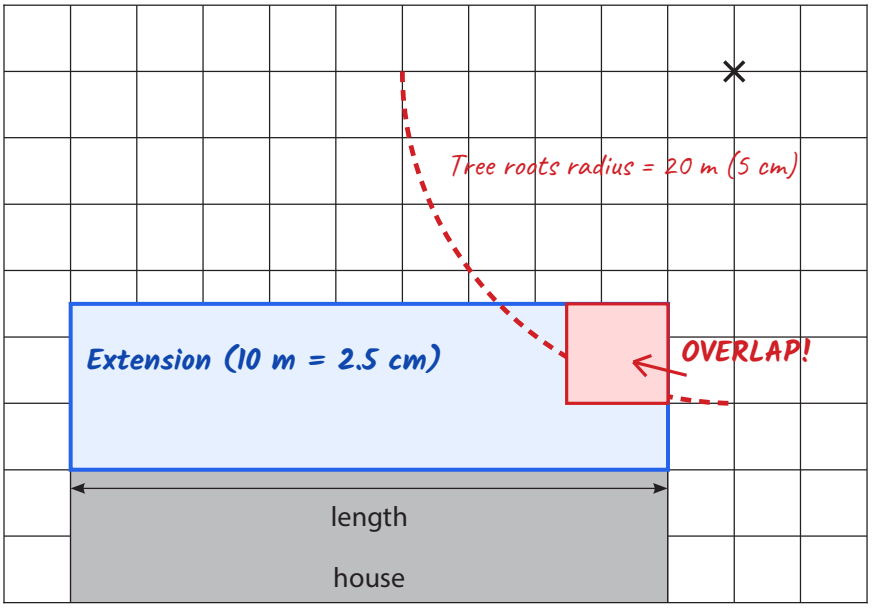
**6% decrease** %

**(Total for Question 5 is 5 marks)**

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**6** Kasim wants an extension built onto his house.

The diagram shows part of the house and the centre of a tree.




**Key** 1 cm on the grid represents ..... 4 m on the ground
✕ represents centre of tree

The length of the house is 36 m.

(a) Complete the key.

**(1)**

House = 9 cm on grid = 36 m --> 36 m ÷ 9 cm = 4 m per cm (1 cm = 4 m) [A1]

The extension will be in the shape of a rectangle with width of 10 m and the same length as the house.

The roots of the tree grow in a circular shape.
The roots grow to a maximum length of 20 m from the centre of the tree.

The extension will need deep foundations if built over the roots.

(b) Will the extension need deep foundations?
Use the grid to show why you think this.

**(2)**

- Extension: 10 m ÷ 4 = 2.5 cm out from house
- Tree roots: 20 m ÷ 4 = 5 cm radius; overlaps extension by > 1.5 cm [M1]

Yes (overlap > 1.5 cm) [A1]

**(Total for Question 6 is 3 marks)**

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**7** Tess works for a drinks company.
She is told that it is better for the environment if less metal is used when making cans.

She starts to design a new can that is taller and narrower than the old can.
The new can

- is in the shape of a cylinder
- has a diameter of 52 mm
- has a height of 154 mm
- holds the same amount of drink.

Tess knows that 27 332 $mm^2$ of metal is used to make the top, the side and the bottom of each old can.

She thinks that her design will use less metal to make the top, the side and the bottom of each new can.

Is Tess correct?
Show why you think this.

**(4)**

Step 1: Radius of new can = 52 ÷ 2 = 26 mm

Step 2: Area of 2 circular ends = 2 × pi × $r^2$

**2 × 3.14 × $26^2$ = 2 × 3.14 × 676 = 4245.28 $mm^2$ [M1]**

Step 3: Curved surface area = pi × d × h

**3.14 × 52 × 154 = 3.14 × 8008 = 25 145.12 $mm^2$ [M1]**

Step 4: Total surface area = 4245.28 + 25 145.12 = 29 390.4 $mm^2$ [M1]

(or 29 405 $mm^2$ using pi on calculator)

**Conclusion: No, Tess is not correct.**

The new can uses 29 390.4 $mm^2$ of metal, which is MORE than the old can (27 332 $mm^2$). It uses 2058.4 $mm^2$ extra metal! [A1]

**No (29 390.4 $mm^2$)**

**(Total for Question 7 is 4 marks)**

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**8** Last week 263 people passed their driving test at a test centre.

The table shows information about the number of driving tests these people took before passing.

| Number of driving tests taken | Frequency |
|---|---|
| 1 | 118 |
| 2 | 74 |
| 3 | 43 |
| 4 | 21 |
| 5 or more | 7 |

(a) What is the probability that a person chosen at random passed their driving test on the first attempt?

**(1)**

Total candidates = 118 + 74 + 43 + 21 + 7 = 263 (given)

Frequency for 1 attempt = 118

**P(1st attempt) = 118 ÷ 263 = 118/263 [A1]**

**118/263**

(b) What is the probability that a person chosen at random did not pass their driving test on the first attempt?

**(2)**

People who did NOT pass on 1st attempt: 263 – 118 = 145 [M1]

(or sum of other rows: 74 + 43 + 21 + 7 = 145)

**Probability = 145 ÷ 263 = 145/263 [A1]**

**145/263**

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Keely sees a car she wants to buy.
The purchase price of the car is £4200

She can pay the purchase price in full or use this payment plan.

**Payment plan**

Pay $\frac{2}{5}$ of purchase price today

24 monthly payments of £112.90

Keely knows that using the payment plan will cost more than paying in full.

(c) How much more will Keely pay using the payment plan?

**(3)**

*Step 1: Calculate deposit paid today (2/5 of £4200)*

£4200 ÷ 5 = £840

Deposit = £840 × 2 = £1680 today [M1]

*Step 2: Calculate total of 24 monthly payments*

24 × £112.90 = £2709.60 [M1]

*Step 3: Total cost of payment plan*

£1680 + £2709.60 = £4389.60

*Step 4: Difference between payment plan and full price*

£4389.60 − £4200.00 = £189.60 [A1]

£ 189.60

**(Total for Question 8 is 6 marks)**

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**9** Ryan is doing a project about sugar at school.
He wants to compare the amount of sugar in grapes with the amount of sugar in cookie dough.

Ryan finds this information

- grapes weighing 92 g contain 15 g of sugar
- cookie dough weighing 610 g contains 110 g of sugar.

Ryan thinks that there is a higher percentage of sugar in the cookie dough than in the grapes.

Is Ryan correct?
Show why you think this.

**(3)**

*Step 1: Calculate percentage of sugar in grapes*

$(15 \div 92) \times 100\% = 0.16304... \times 100\% = 16.30\%$ [M1]

*Step 2: Calculate percentage of sugar in cookie dough*

$(110 \div 610) \times 100\% = 0.18032... \times 100\% = 18.03\%$ [M1]

*Step 3: Compare percentages*

$18.03\%$ (cookie dough) $>$ $16.30\%$ (grapes)

*Alternative: In 610 g of grapes, sugar = 610 × (15 ÷ 92) = 99.46 g < 110 g*

**Conclusion: Yes, Ryan is correct.**

*Cookie dough has 18.0% sugar, which is higher than grapes at 16.3% sugar. (18.03% > 16.30%). [A1]*

Yes (18.0% > 16.3%)

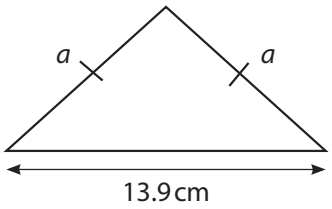
**(Total for Question 9 is 3 marks)**

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**10** Here is an isosceles triangle.

It has a

- base length of 13.9 cm
- perimeter of 38.2 cm.




(a) Work out the length of side *a* on the triangle.

**(2)**

Step 1: Subtract base from perimeter to find total of the 2 equal sides

$2a = 38.2 - 13.9 = 24.3$ cm [M1]

Step 2: Divide by 2 to find length of side a

$a = 24.3 \div 2 = 12.15$ cm [A1]

$a =$ 12.15 cm

Here are some calculations.

| | | | |
|---|---|---|---|
| $40 \div 14$ | $40 - 10 - 10$ | $14 \times 38$ | $\frac{40 - 14}{2}$ ✓ |
| $38 - 14 \times 2$ | $\frac{40 + 14}{2}$ | $38 - 14 - 14$ | |

☑ (b) Which calculation is suitable as a check using estimation?
Tick [✔] the calculation to show your answer.

**(1)**

Check explanation:

Round perimeter 38.2 cm to 40 cm, and base 13.9 cm to 14 cm.

Estimated side $a = (40 - 14) \div 2 = 26 \div 2 = 13$ cm $\approx 12.15$ cm. [B1]

**(Total for Question 10 is 3 marks)**

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**11** (a) Write 6.8% as a decimal.

**(1)**

6.8% = 6.8 ÷ 100 = 0.068 [A1]

0.068

Jess invests £3800 into a savings account for 3 years.

She will not put any extra money into the account.
She will not take any money out of the account.

The investment will earn 2.4% compound interest per year.

(b) Work out the total amount of interest earned after 3 years.

**(4)**

*Step 1: Interest rate and multiplier*

Interest rate = 2.4% per year

Multiplier = 1 + (2.4 ÷ 100) = 1.024 [M1]

*Step 2: Calculate year-by-year balance*

End of Year 1: £3800 × 1.024 = £3891.20

End of Year 2: £3891.20 × 1.024 = £3984.5888

End of Year 3: £3984.5888 × 1.024 = £4080.2189... [M1]

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Step 3: Total balance after 3 years

Total value = £4080.22 (rounded to 2 d.p.)

Step 4: Total interest earned = Total balance – Initial investment

Interest = £4080.2189... – £3800.00 [M1]

= £280.2189... ≈ £280.22 [A1]

Alternative single calculation:

£3800 × $(1.024)^3$ = £4080.2189...

£4080.22 – £3800 = £280.22

£ 280.22

**(Total for Question 11 is 5 marks)**

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**12** Andy works at a medical centre.
He reads a news report about a shortage of doctors in some parts of the UK.

The report states the following figures for the UK

- the average number of patients per doctor is 1734
- the range is 826 patients per doctor.

The table shows some information about the number of patients for each doctor at the medical centre.

| Doctor | Number of patients |
|---|---|
| A | 1348 |
| B | 1847 |
| C | 1760 |
| D | 1562 |
| E | 1240 |
| F | 1703 |

Andy needs to write a report to compare the figures for the medical centre with the figures for the UK.

He needs to comment on

- a comparison of the average number of patients per doctor
- the consistency of the number of patients per doctor.

Write comments for the report.
Remember to use calculations and figures to support your comments.

**(6)**

*Step 1: Calculate the range for the medical centre*

**Highest = 1847 (Doctor B) , Lowest = 1240 (Doctor E)**

**Range = 1847 – 1240 = 607 patients per doctor [M1 A1]**

*Step 2: Calculate the mean average for the medical centre*

**Total = 1348 + 1847 + 1760 + 1562 + 1240 + 1703 = 9460 [M1]**

**Mean = 9460 ÷ 6 = 1576.67 ≈ 1577 patients per doctor [A1]**

*(Turn over for full comparative report comments)*

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Summary of comparative figures:

- **Medical Centre Mean = 1577 (Median = 1632.5) vs UK Mean = 1734**
- **Medical Centre Range = 607 vs UK Range = 826**

## Comment 1: Comparison of Average Number of Patients

The average number of patients per doctor at the medical centre (mean of 1577, or median 1632.5) is LOWER than the UK national average of 1734 patients per doctor (157 fewer patients per doctor). [A1]

## Comment 2: Comparison of Consistency (Range)

The range of patients per doctor at the medical centre (607) is SMALLER than the UK national range (826), meaning that doctor caseloads at the medical centre are MORE CONSISTENT than nationally. [A1]

## Report for Medical Centre Management:

1. Average: Medical centre mean is 1577 patients per doctor, lower than the UK average of 1734 patients per doctor (difference of 157 patients).

2. Consistency: Medical centre range is 607 patients per doctor, narrower than the UK range of 826 patients per doctor, showing more consistent doctor caseloads.

**Both comments fully supported by calculations: Mean = 1577 & Range = 607. [6/6]**

**(Total for Question 12 is 6 marks)**

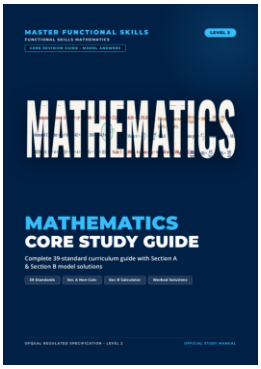
**TOTAL FOR SECTION B IS 48 MARKS**
**TOTAL FOR PAPER IS 64 MARKS**

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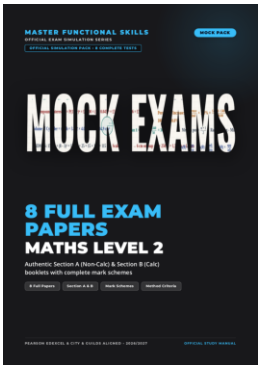



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